# How does the Forcing Profile and Power Extraction from Vertical Axis and Horizontal Axis Wind Turbines Differ Under Equal Frictional Conditions?

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## Plagiarism declaration

> This piece of work is a result of my own work and I have complied with the Department's guidance on multiple submission and on the use of AI tools. Material from the work of others not involved in the project has been acknowledged, quotations and paraphrases suitably indicated, and all uses of AI tools have been declared.

I, **Jacob McKerrell** confirm that I have read, understood, and have adhered to the plagiarism declaration above.

## AI Usage Declaration

- Gemini was used to find papers discussing some of the covered topics (Gibbs Phenomenon and Thrust double integral)
- Gemini corrected my understanding of the blade swept area (S). It clarified that S only needed to be considered as the 1-dimensional cross section, equal to the $z$ component of the area, and not a 2D area
- Co-pilot generated my github commit messages

## Link to GitHub Codebase

All codes used to produce the analysis and results in this report can be found at the GitHub repository: https://github.com/Jacob-McKerrell/comp_math_individual.

# 1 Forcing Profile of Vertical Axis Wind Turbines

To approximate the forcing profile of a Horizontal Axis Wind Turbine (HAWT), a Gaussian model is used in both the $x$ and $z$ directions. Maximum air velocity decrease occurs at the centre of the turbine hub and decays as the distance from the hub increases. For a Vertical Axis Wind Turbine (VAWT), the Gaussian forcing profile in the $x$ direction is maintained, since the forcing will still decrease as the horizontal distance from the hub increases. However, for the $z$ direction, a different model, which more accurately represents the turbine's behaviour on the surrounding flow, is required. Gyromills are a type of VAWT which sweep out a cylindrical shape of air. A uniform forcing profile in the $z$ direction is a suitable, though simplified model for this type of turbine [8]. However, directly using a top hat function will cause issues, since a function of this form will be subject to the Gibbs phenomenon [3]. For any piecewise smooth function, there are errors around discontinuous points when differentiating and attempting to calculate the Fourier transformation. These errors, visible as an over and undershooting sine wave around the maximum amplitude point, have a magnitude proportional to the jump size [3]. To combat this issue, we will modify our vertical forcing profile $(z)$ to be:

$$\tanh\frac{z-a}{\delta} - \tanh\frac{z-b}{\delta} \tag{1}$$

where $b-a$ is the height of the wind turbine blades. Since $\tanh\left(\frac{x}{\delta}\right)$ tends towards the step function as $\delta \to 0$, equation 1 will converge towards the top-hat function when we reduce the value of $\delta$. The desired uniform property is now successfully maintained (shown in Figure 1), while also being differentiable at all points. Due to this smooth and continuous nature, the Fourier Transform is now possible without the unwanted Gibbs Phenomenon taking place.

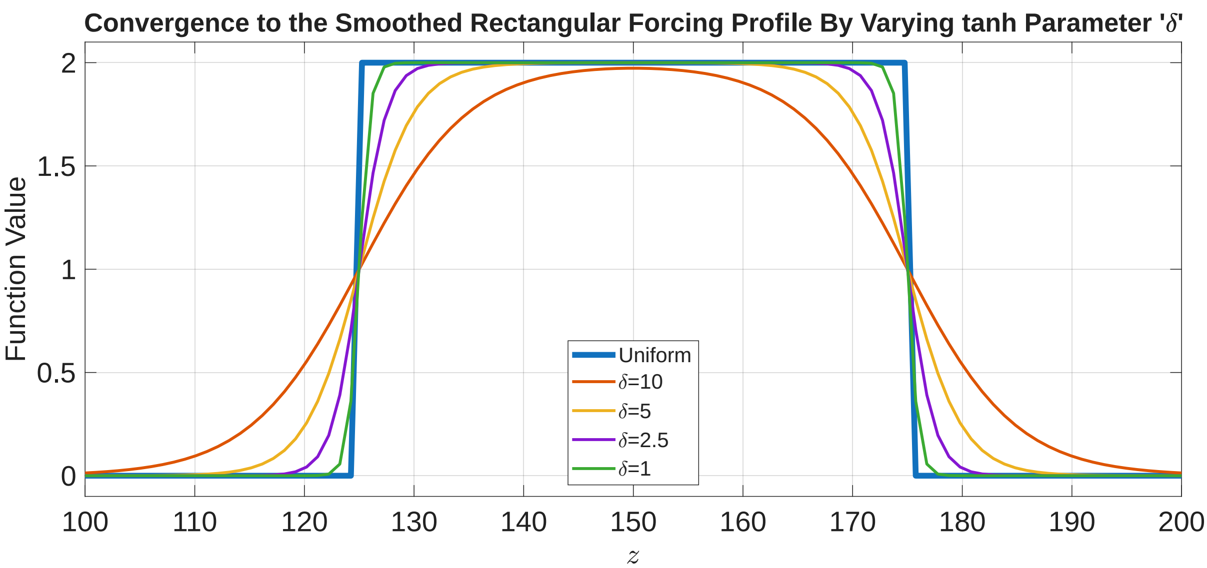



Figure 1: Using $\tanh\frac{z-125}{\delta} - \tanh\frac{z-175}{\delta}$ as an estimator for the top hat function

Our body force $f_x$ is now modelled by:

$$f_x(x,z) = A\ \exp\left(-\frac{(x-x_0)^2}{2\sigma_x^2}\right)\cdot\left(\tanh\frac{z-a}{\delta} - \tanh\frac{z-b}{\delta}\right) \tag{2}$$

## 2 Mathematical Modelling and Power Calculations

To fairly compare the two turbine models, we must ensure identical total momentum extraction from the flow. We equate the total frictional forces applied by the turbine on the wind, which is equivalent to the total thrust acting on the turbine (by Newton's third law). We integrate the distributed forces over the full area domain to find the thrust[9]:

$$T = \int_{L_z} \int_{L_x} f_x(x, z) dx dz \tag{3}$$

Before normalisation, the friction acting on the flow in the VAWT was less than for the HAWT. By increasing the forcing amplitude($A_{VAWT}$) the strength of the VAWT can be increased to match that of the HAWT. A fair comparison of turbine efficiencies can now occur. By numerically approximating $T_{HAWT}$ and $T_{VAWT}$ using Matlab's trapz function [4] we can use these values to calculate the appropriate forcing amplitude for the VAWT turbine such that $T_{HAWT} = T_{VAWT}$.

$$A_{VAWT} = A_{HAWT} \cdot \frac{T_{HAWT}}{T_{VAWT}} \tag{4}$$

Recall Equation (3) from our group project:

$$U_0 \left( \frac{\partial^2 w}{\partial z^2} + \frac{\partial^2 w}{\partial x^2} \right) = -\frac{\partial f_x}{\partial z} \tag{5}$$

When working with our Gaussian-based HAWT model, we used the method of finite differences to calculate $\frac{\mathrm{d}}{\mathrm{d}z}\hat{f}_k$. Applying this method to the tanh top hat function approximation would result in a loss of our desired continuous property, risking the return of the Gibbs Phenomenon. Instead, we will find the analytical derivative of the VAWT $f_x$(2) before its transformation to $\hat{f}_k(z)$. Once this has been transformed into the Fourier space, we can calculate $u$ and $w$ using the same spectral method we used for the HAWT. The RHS (analytical derivative of $f_x$) of Equation (5) is as shown:

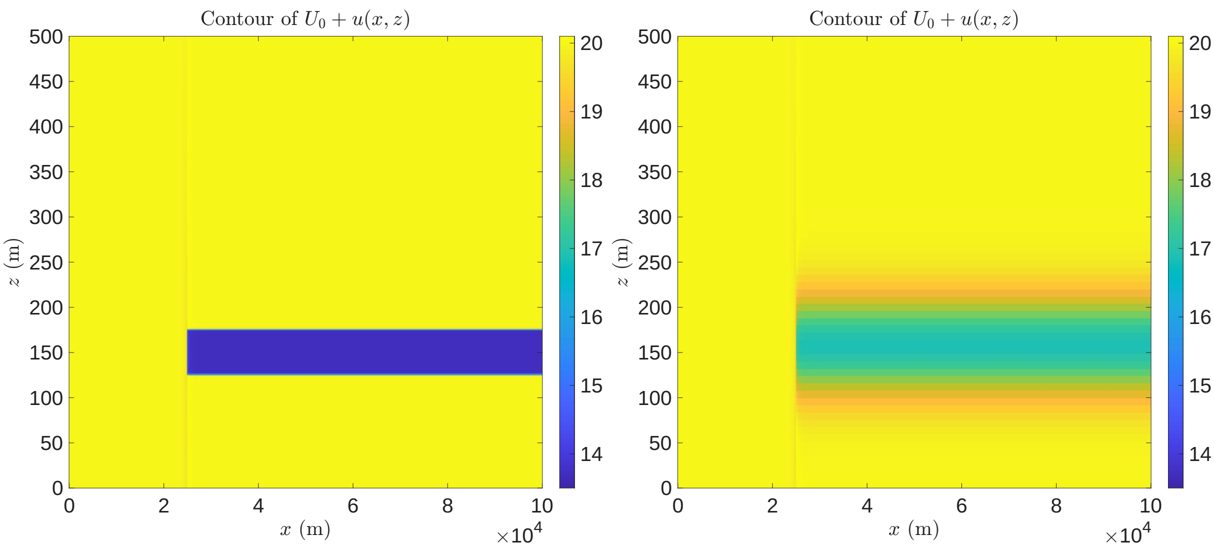
$$-\frac{\partial f_x}{\partial z} = A_{VAWT} \exp\left( -\frac{(x - x_0)^2}{2\sigma_x^2} \right) \cdot \frac{1}{\delta} \left( \operatorname{sech}^2\left( \frac{z-a}{\delta} \right) - \operatorname{sech}^2\left( \frac{z-b}{\delta} \right) \right) \tag{6}$$

Power is fundamentally defined as the product of force and velocity. Since the forcing profile $f_x(x, z)$ and the horizontal velocity matrix $u(x, z)$ are both 2-dimensional matrices, the power extracted by the turbine, which is equal in magnitude to the energy per unit time lost by the flow of air, will be calculated as the integral of the product of these two matrices, over the entire area domain. The total kinetic energy in the wind per unit time is given by $P = \frac{1}{2}\rho S U_0^3$[6]: $U_0$ is the background wind speed; $\rho$ is air density ($\sim 1.20769 kg/m^3$ at 150m [11]); and S is the blade swept area. Because we are only measuring the momentum loss in the $x$ direction, the area is the cross-section in the $z$ direction. Using the height of the VAWT $(b - a)$, ensures each turbine is subject to the same area of air. To calculate the efficiency of the turbines, we calculate the power coefficient, which is the ratio of the power extracted by the turbine against the total power in the wind [6]:

$$C_p = \frac{P_{turbine}}{P_{wind}}[6] \rightarrow \frac{\int_{L_z} \int_{L_x} f_x(x, z) \cdot (U_0 + u(x, z))\, dx dz}{\frac{1}{2}\rho S U_0^3} \tag{7}$$

# 3 Analysis of Results

We have modeled the VAWT to have 50m long blades centered at 150m ($Z_0$). We have increased strength of the VAWT through the normalisation (4) of the forcing amplitude to $A_{VAWT} =\sim 0.5013m/s^2$. This turbine will now impose the same amount of friction on the surrounding air as the HAWT with $A_{HAWT} = 0.5m/s^2$. The heatmaps below show the wake behind each of the turbines for $\delta = 1$. The HAWT spans a larger area and the amplitude of velocity reduction decreases gradually from about $3.2m/s$ at the turbine hub, to $\sim 0m/s$ as this distance increases. Conversely, the strong uniform forcing profile of the VAWT, maintains a high $\sim 6.3m/s$ velocity reduction within the acting height of the wind turbine which almost immediately drops to zero outside of this height range.




(a) VAWT wake heatmap

(b) HAWT wake heatmap

Figure 2: Heatmaps comparing the wakes of HAWT and VAWT

We calculated the HAWT efficiency to be 24.59% whilst the VAWT was a much higher 43.93%. The heatmap show that the VAWT is more effective at slowing down the wind across the whole acting area, whilst the HAWT has a narrower peak velocity reduction region. A larger percentage of the energy in the oncoming wind is extracted in the top-hat model compared to the Gaussian model. Betz limit[1] describes the theoretical maximum turbine efficiency (regardless of mechanism) to be $\frac{16}{27}$. It is impossible to extract more than $\sim$ 59.3% of the kinetic energy from the wind. Both of the efficiency calculations fall below this upper limit. However, due to simplifications in the model, these results don't perfectly match reality. For example, the VAWT efficiency has been overestimated due to the lack of consideration for backtracking. VAWTs fall vulnerable to backtracking due to the blades moving parallel to the wind. They need to do work against the flow of wind at points throughout their full rotation [5]. The current uniform estimation doesn't account for this phenomenon, since the drag experienced in the retreating direction is ignored. Perpendicular blades protect HAWTs from this issue. Their design allows energy to be extracted through the full rotation of the blades when placed under consistent wind flow [5]. Overall, whilst VAWT can operate better in high wind turbulence, wind fluctuations, and high directional variability [10], wind power extraction efficiency in HAWT is often better than that of VAWT[2]. HAWTs remain more common due to their mature technology and long record of operational success [7].

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